MIKHAIL CHULOSHNIKOV

SENIOR GOLANG/PHP DEVELOPER | ENGINEERING TEAM LEAD | HIGH-LOAD ARCHITECT | SYSTEMS SECURITY RESEARCHER

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B2B contract

Senior Golang/PHP Developer & Engineering Lead with 17+ years of experience in architecting high-concurrency systems and resilient infrastructures. Specialist in **Golang**, distributed microservices, and performance optimization for platforms handling **1B+ records**.

I combine a **"low-level first"** mindset (C/C++, memory corruption research, custom network protocols) with modern leadership practices, having a proven track record of scaling engineering teams from scratch and **boosting velocity by 50%**. Expert in building mission-critical ecosystems where security, zero-downtime, and extreme throughput are non-negotiable. Whether it's writing custom Go extensions or orchestrating global banking platforms, I engineer systems that are secure by design and scalable by default.

ACCOMPLISHMENTS

Fintech & Core Banking

- **Core Engine Architecture:** Architected and built a high-load processing engine for FX, multi-currency routing, and payment gateways from the ground up.
- **Banking Infrastructure Launch:** Led the end-to-end creation of a banking ecosystem, including core service design, team assembly, and business process automation.
- **Compliance & Security:** Implemented secure payment schemes and protocols (SWIFT, SEPA) ensuring full regulatory compliance.

Scale & Performance

- **High-Load Data Management:** Managed a cluster of 6+ microservices handling **1 Billion+ data records** with high availability.
- **Engineering Management:** Reorganized development processes and team structure, increasing team velocity from **70% to 120%** using a modified Scrum framework.
- **System Optimization:** Optimized notification systems and DB interactions for a platform with **tens of millions of active users**.

Systems Research & Security

- **Vulnerability Research:** Conducted deep-dive analysis of browser engines and PDF readers; developed PoC RCE exploits for memory corruption flaws (WinDbg, x86 Assembly, heap-spraying).
- **Low-Level Engineering:** Engineered ultra-lightweight (4KB) remote management agents and infrastructure using C/WinAPI and Lua for extreme stealth and efficiency.

- **Code Protection:** Developed a polymorphic code protector with a custom translator (Python/JS to pseudocode) to bypass heuristic analysis.

Zero-to-One Infrastructure

- **Infrastructure from Scratch:** Built the entire IT department and business processes for a banking platform (team, environments, CI/CD, documentation).
- **E-commerce Foundations:** Designed billing systems and search engine architecture for a major platform during its hyper-growth phase.
- **Decentralized Systems:** Developed trust-ring networks and anti-censorship systems with a focus on decentralized security.

PROFESSIONAL EXPERIENCE

Team lead, Aventus IT **Jul 2024 to Oct 2025**
Almaty

- **Engineering Management:** Leading and mentoring a backend development team of 7 people.
- **Business Process Organization:** Establishing and streamlining internal workflows and cross-departmental operations.
- **System Architecture:** Designing robust architectures for complex projects and high-performance services.
- **CI/CD Excellence:** Integrating and optimizing continuous integration and delivery pipelines to accelerate deployment.
- **Core Development:** Direct contribution to large-scale flagship projects and their interconnected microservices ecosystem.

Project lead / Senior developer, Silverbird (Theropod LTD) **Oct 2022 to Mar 2024**
London

- **Core Banking Development:** Engineering of core banking engines and all related financial infrastructure.
- **Multi-PSP Integration:** Design and development of systems for seamless multi-PSP (Payment Service Provider) integration.
- **Payment Processing Core:** Architecture and development of the core engine for Inbound/Outbound payments, FX transactions, and intelligent payment routing.
- **Leadership & Stakeholding:** Acting as Team Lead and Stakeholder for the core banking and payment modules.
- **External Ecosystems:** Integration of external PSPs and development of specialized Beneficiary Modules.
- **Security Auditing:** Conducting comprehensive security audits for financial components.
- **Banking Domain Expertise:** Full-cycle implementation of payments, FX, and transactions. Deep integration with payment schemes: **SWIFT, SEPA, Faster Payments, T2**, and platforms like **TUUM / Integrated Finance**.

Senior developer, MTS (Mobile TeleSystems) **Jul 2022 to Sept 2022**
Moscow

- **Bitrix Development:** Engineering and implementation of new functional modules within the Bitrix framework.
- **Legacy Migration:** Architectural transition and migration of legacy Bitrix functionality to new microservices using a modern technology stack.

Team lead, TSUM (Mercury Group) **Apr 2021 to Apr 2022**

Moscow

- **Cross-functional Leadership:** Managed a complete product cell including Developers, QA, Product Managers, and Business Analysts.
- **Massive Scale:** Managed High-Load systems handling nearly **1 Billion data records**.
- **Service Ownership:** Full accountability for **6+ mission-critical services** within the e-commerce ecosystem.
- **Marketing-Tech Strategy:** Spearheaded the Marketing product-engineering bridge, acting as a primary Stakeholder.
- **Team Transformation:** Completely rebuilt and upskilled the team to align with modern product requirements and tech standards.
- **Innovation Driver:** Initiated and oversaw the development of new services using cutting-edge technologies to solve specific business needs.
- **Process Excellence:** Built internal development workflows from scratch and streamlined interaction protocols between business units and external teams.

Head of Development, Silverbird (Theropod LTD) **Sept 2020 to Apr 2021**

London

- **Banking Platform:** Orchestrated the end-to-end creation and implementation of the core banking platform, infrastructure, and essential services.
- **Operational Leadership:** Assembled the engineering team from scratch; established business processes, administration protocols, and organizational workflows.
- **Strategic Partnership:** Managed cross-company interactions with partners, including joint strategic planning, training, and deep integration.
- **Project Launch:** Designed and launched ~10 mission-critical projects from the ground up, including API services, microservices, and integration layers.
- **Macro-Banking Integration:** Audited and deployed multi-level macro-banking platforms across diverse technology stacks, ensuring seamless platform enhancements.
- **Update Architecture:** Designed complex technical update flows and global architecture for application components and infrastructure.
- **Technical Governance:** Developed comprehensive technical documentation and administered both staging and production environments.

Tech analyst / Senior developer, Gosuslugi **Sept 2020 to Apr 2021**

Moscow

- **Feedback System:** Development of the core feedback subsystem and related services.
- **Fault Tolerance:** Deep optimization and profiling to ensure system stability under extreme high-load conditions.
- **System Refactoring:** Structural refactoring of core system components for better maintainability and performance.
- **System Integration:** Orchestrating complex integrations with both internal and external third-party systems.
- **State Auth Integration:** Implementation of transparent SSO and authorization via ESIA (National Auth System).
- **Data Normalization:** Bringing complex data schemas to normal forms to ensure data integrity and performance.

- **ROI Integration:** Integration with ROI (Return on Investment) tracking and analytical systems.

Senior developer, TSUM (Mercury Group) **Feb 2018 to Jul 2020**
Moscow

- **Internal Product Development:** Engineering and maintenance of the company's core internal ecosystem and software products.
- **Data Synchronization:** Debugging and fine-tuning complex data synchronization processes between various system components.
- **Search Engineering:** Implementation of high-performance search functionality and all related search-driven features.
- **Service Innovation:** Developed and launched several new services, including product feed generators, analytical monitoring systems, link shorteners, and QR-code generation engines.
- **MarTech Evolution:** Continuous development and scaling of marketing-focused services and promotional tools.

Team lead, JivoChat (JivoSite) **Apr 2016 to Nov 2017**
Moscow

- **Engineering Leadership:** Managed a team of 6 people, overseeing architectural planning, code quality, and hands-on development for mission-critical business projects.
- **Process Transformation:** Implemented Scrum (tailored to team needs) including estimation, forecasting, and retrospectives. Automated major parts of the system to increase operational transparency.
- **Efficiency Breakthrough:** Boosted team productivity from **70% (missed deadlines) to 120% (over-delivery)** through process optimization and workflow alignment.
- **Stakeholder Management:** Orchestrated communication between IT and other departments, ensuring a unified information field across all teams and stakeholders.
- **Release Engineering:** Owned the full release management cycle and deployment processes.
- **Team Growth & Mentorship:** Reorganized the team structure, leading to a significant increase in both individual and collective competencies.
- **Knowledge Sharing:** Achieved team interchangeability through cross-training; organized and led internal seminars, conferences, and technical workshops.

Senior developer, VK (Vkontakte) **Jan 2016 to Mar 2016**
Moscow

- **NDA Projects:** Development for high-load specialized projects under strict NDA.

Senior developer, Activizm.ru **Aug 2014 to Dec 2015**
Moscow

- **Feature Engineering:** Developed new core features and provided comprehensive support for the existing platform functionality.
- **AdTech Integration:** Seamlessly integrated Google AdWords and Yandex.Direct APIs for automated marketing management.
- **Data Syndication:** Managed large-scale data exports to Google Merchant Center (Content Center) for product listing ads.
- **Marketing Ecosystem:** Engineered product feed generation systems and integrated them with various external marketing services.

- **Performance Engineering:** Conducted deep code refactoring, system profiling, and database optimization to improve platform stability.

Senior developer, Mamba **Nov 2013 to Jul 2014**
Moscow

- **High-Load Engineering:** Developed features and core functionality for a massive social platform serving tens of millions of active users.
- **Notification Engine:** Designed and implemented a high-performance notification delivery system.
- **Low-Level Integration:** Integrated and debugged system interactions with custom-built C-based daemons to ensure low-latency processing.
- **Mail Service Refactoring:** Architected a complete refactor of the email interaction system and mail gateway protocols.
- **Abuse & Subscription Logic:** Enhanced the processing systems for user complaints, reports, and automated unsubscription flows.
- **Database Performance:** Optimized complex database operations utilizing both SQL and RRDtool (Round Robin Database) for time-series data.

Senior developer, Enter.ru **May 2012 to Aug 2013**
Moscow

- **Core Architecture:** Development of the core information infrastructure and engineering of external APIs.
- **Fintech Integration:** Integrated credit brokerage systems to enable real-time online installment and credit purchases.
- **Enterprise Systems:** Orchestrated deep integration with 1C enterprise management systems.
- **Marketing Automation:** Integrated the Ofsys mailing system for automated customer communications.
- **CMS & Extensions:** Deployed WordPress as a CMS solution, integrated it into the main site, and developed numerous custom plugins.
- **Legacy Decommissioning:** Performed a comprehensive site refactoring, including the architectural migration to remove Symfony dependencies.

Senior developer, Wikimart **Aug 2009 to Apr 2012**
Moscow

- **Billing Infrastructure:** Designed and developed the core billing system from the ground up.
- **Database Engineering:** Performed comprehensive refactoring of database trigger systems and complex stored procedures.
- **Data Ingestion Services:** Engineered high-throughput services for importing **YML and CSV** data feeds containing massive product catalogs from various merchants.
- **Search & Merchandising:** Developed the site-wide search engine and built merchandising systems (promotions, special offers, and virtual storefronts).
- **Growth Tools:** Developed the affiliate program platform and contributed to the internal **CRM system** development.
- **Full-Stack Evolution:** Led the total site redesign and participated in global system architecture planning, delivering numerous full-stack features.

Senior developer, MIF **Sept 2008 to Aug 2009**
Moscow

- **Full-Chain Vulnerability Analysis:** Conducted end-to-end security research on major browser engines and Adobe Reader. Developed **functional RCE (Remote Code Execution) exploits (Only Proof-of-Concept)** to demonstrate critical memory corruption flaws using **WinDbg, x86 Assembly, and advanced heap-spraying techniques**.
- **High-Concurrency C&C Architecture:** Engineered ultra-lightweight remote management agents (4KB binaries) and command-and-control (C2) infrastructures using **C (WinAPI) and Lua**. Focus on extreme binary optimization and stealthy network communication.
- **Polymorphic Code Protection:** Developed a sophisticated machine-based obfuscator. It featured a custom code translator (Python/JS to pseudocode) designed to bypass heuristic analysis and static signature detection engines.
- **Algorithmic Network Topology:** Designed and implemented automated systems for large-scale content generation and inter-linked network structures (Ring & Pyramid topologies). Optimized link-graph algorithms to maximize search engine indexing efficiency.
- **Infrastructure Monetization:** Developed specialized platforms for client hardware resource acquisition and satellite site generation systems.
- **Systems Mindset:** This era defined my "low-level first" approach. Understanding memory corruption and CPU-level execution is what allows me today to build high-load systems that are inherently secure.

CURRENT VENTURE

Alunea (Founder & Lead architect) **alunea.io**

An advanced learning ecosystem that bridges the gap between passive content consumption and active speaking. Alunea transforms streaming platforms into personal classrooms using AI-driven subtitles and voice-interactive tutors.

- **AI Voice Tutor "Violet":** Interactive AI calls for speaking practice, utilizing saved vocabulary from shows for active recall.
- **Real-time AI Subtitles:** Context-aware dual subtitles and on-the-fly generation for Netflix, YouTube, and Disney+.
- **Unified Ecosystem:** Instant word-sync between browser extension, mobile apps (iOS/Android), and AI backend.
- **Adaptive Learning:** System analyzes skill levels to tailor conversation difficulty and recommendation logic.

EDUCATION

Moscow University of Industry and Finance "Synergy" **Sept 2022 - Jan 2026**
Bachelor’s Degree, Fullstack Development, Moscow

Modern Technical Institute **Sept 2012 - Aug 2015**
Master’s Degree, Service, Ryazan

Moscow College of Mathematics **Sept 2004 - Aug 2008**
Associate Degree, Computer Systems Engineering and Software Development, Moscow

TECHNICAL PROFICIENCIES

Primary languages:	Golang, PHP, Python
Secondary languages:	Java, JavaScript, Typescript, Lua, Gdscript, C, C++, ASM, Tcl, Delphi, VBA, Bash
Data & Storage Systems:	MySQL, PostgreSQL, MS SQL, Interbase, T-SQL-92, ClickHouse, Cassandra, Redis, Memcached, Firebase, Kyoto, Database Replication, Database Sharding, Database Partitioning, Performance Tuning, Database Profiling
Search engines:	Elasticsearch, Sphinx, Solr
Message brokers:	RabbitMQ, Kafka
System Architecture & High Load:	Distributed Systems, Microservices, Load Balancing, Zero-Downtime Altering, Concurrency, Enterprise Service Bus
Frameworks & Libraries:	Symfony, Laravel, Yii, Zend Framework, Doctrine, Spring Boot, Spring Security, Spring Social, Hibernate, Django, SQLAlchemy, Boost, Vue.js, Angular, Vuetify, Webpack, Electron, Ionic, Capacitor, Custom Extensions (C)
Security & Systems Research:	0-day Exploit Development, Remote Code Execution (RCE), XSS, Buffer Overflow, Cryptors, Shellcode, Polymorphic Code Protection, Reverse Engineering, Binary Patching, Deobfuscation, Import Table Recovery, Security Audits
Low-Level & OS Internals:	WinAPI, Kernel Module Patching, WinDbg, OllyDbg, GDB, Strace, Raw Sockets, Packet Inspection (TCP/UDP), Stack/Heap/Registers Analysis, COM/RPC, Binary Optimization
Infrastructure & DevOps:	Docker, Ansible, GitHub Actions, Nginx, Caddy, Apache HTTP Server, Tomcat, Jetty, Grafana, Loki, Alloy, Linux, Git, SVN, Mercurial, Jira, Confluence, Redmine, Trac
Algorithms & Structures:	B-Tree, Red-Black Tree, Binary Search Tree, QuickSort, Bubble Sort, Shaker Sort, Binary Search, Hash Tables, Queue, Deque, Stack, Singly Linked List, Doubly Linked List, System V IPC: Shared Memory, Multithreading, Graph Theory
Patterns & Other:	SOLID, GoF, Undetermined State Machines, Trust Ring Architecture, Game/World Physics, Decorator, Singleton, Proxy, Outbox, Factory, Abstract Factory, Registry, Command, MVC, Prototype, Adapter, Facade, Workflow, KIS, Fork, UML, Late Static Binding, FIFO, LIFO, IoC, DI

PUBLICATIONS

Buffer Overflow for Beginners - Technical deep-dive into memory corruption, stack smashing, and the fundamentals of writing exploit code.
ARP Spoofing: Network Vulnerability - Analysis of fundamental local network weaknesses and practical demonstration of MITM (Man-in-the-Middle) concepts.
The Power of UDF in MySQL - Extending MySQL capabilities with C and OpenSSL to implement robust field-level data encryption.
File-based hash map - Using PHP file hash map as persistent data storage.

CODE PORTFOLIO

C / C++:

C/C++/Golang: Algorithms
C++: MySQL OpenSSL Plugin
C++: ARP Spoofer
C++: Shellcode Generator
C++: YansDB (NoSQL Storage)
C++: TCP/IP Broadcast Server
C++: Lightness HID Tool

Go, Python, Lua:

Go: LSM-tree based storage
Go: Concurrent Crawler
Go: Classes implementation for Go
Python: Helicopter Game
Python: L2 Lottery Bot
Python: Web Cryptor
Python: Lightness Monitor
Lua: Simple Installer
Godot: Fox Wild Hunt

PHP, Java, TypeScript:

PHP: File Hash Map
PHP: Worker Pool (System V)
Java: Email alias - Spring Backend
TypeScript: Email alias - Angular Front
Python: Email alias - Ansible CI/CD
PHP: Various PHP Snippets

LICENSES & CERTIFICATIONS

Zend PHP Certified Engineer **Oct 2018**

LANGUAGES

English: B2 Upper-Intermediate | **Russian:** Native

THE BACKSTORY

Genesis & Early Exploits: Started at 14 with a Delphi book and no internet. By college, I was neck-deep in Pascal, VBA, Firebird, and hacking the lab's WinServer 2x network. My first big project? A WoW server (MaNGOS) on C++ where I learned the pain of patching binary files.

The real fun started with Lineage 2. I built a custom Socks4/5 proxy layer to intercept and modify packets on-the-fly. It was pure raw socket magic completely invisible to anti-cheat systems. Later, I built 0-day exploits for Firefox and Adobe Reader by finding crashes via Google and spending nights in WinDbg/OllyDbg.

The "Grey" Era & Lua Mastery: Developing botnets taught me extreme optimization. I integrated a Lua interpreter into a C-core, stripped the parser, and fed it raw bytecode and the result was a 4KB binary. Try doing that with Python or PHP!

I built my own custom cryptor based on a Python-to-Pseudocode translator with binary tree algorithms and

heavy obfuscation. I even discovered that bypassing Kaspersky back then was possible just by using CMD/FTP for shellcode instead of standard WinAPI - it simply didn't hook the command line the same way.